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## Quantitative Analysis of Polarization Phenomena in CdTe Radiation Detectors

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Polarization phenomena in a Schottky-type CdTe radiation detector were studied. We evaluated the distribution of electric field in a biased CdTe detector by measuring the progressive change of Schottky barrier lowering with time. The parameters of deep acceptors such as detrapping time, concentration, and the depth of the energy level were quantitatively evaluated. In the case of applying the conventional model of charge accumulation, the obtained result shows that the CdTe bulk is never undepleted. We modified the charge accumulation model by taking account of the occupation state of the deep acceptor level. When a modified model is applied, the time that the depletion width in the bulk begins to diminish closely fits the time that the photopeak position begins to shift in radiation measurements. In this paper, we present a distribution of electric field during biasing and a simple method for the evaluation of the parameters of deep acceptors in CdTe bulk.

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### 1. Introduction

Cadmium telluride (CdTe) is a compound semiconductor with a large band-gap energy ( $E_g = 1.47$  eV) and a high atomic number (Cd: 48, Te: 52). Therefore, CdTe offers room-temperature operation and a high absorption efficiency for application as X- and  $\gamma$ -ray detectors. These devices have been used in a wide range of applications, such as medical<sup>1)</sup> and industrial imaging,<sup>2)</sup> and X- and  $\gamma$ -ray astronomy.<sup>3)</sup> The energy resolution of X- and  $\gamma$ -ray detectors is one of the important factors for practical use, but it strongly depends on the leakage current of the detector. To suppress the leakage current, Schottky-type (or diode-type) CdTe detectors have been fabricated.<sup>4,5)</sup> As a result, high energy resolutions such as the FWHMs of the 59.5, 122, and 662 keV photopeaks as small as 1.5, 2.1, and 10 keV, respectively, were obtained from the detectors.<sup>5)</sup>

However, Schottky-type CdTe detectors suffer from an instability with time under bias voltage.<sup>6,7)</sup> For the practical application of CdTe detectors, the time stability of the detectors is one of the most important issues. This instability phenomenon is generally termed as a polarization phenomenon and characterized by a progressive degradation of energy resolution and a shifting of the photopeak toward low energy with time after applying the bias voltage. To our knowledge, the term “polarization” had been used to express the polarization of the space charge distribution which causes the disturbance of electric field in crystal-type detectors.<sup>8)</sup> In the present day, the term “polarization” is used for the time instability phenomena in semiconductor detectors. Earlier works revealed that polarization phenomena result from the accumulation of negative charge on a deep acceptor level during the application of bias voltage.<sup>9–12)</sup> As a result, the distribution of electric field in CdTe bulk varies with time. In fact, the polarization phenomena were observed in several kinds of radiation detectors using compound semiconductors such as HgI<sub>2</sub>,<sup>13)</sup> PbI<sub>2</sub>,<sup>14)</sup> and TlBr.<sup>15)</sup> Deep acceptors can be evaluated using techniques such as deep-level transient spectroscopy (DLTS) and thermally stimulated current (TSC) measurements, but these methods are rather complicated.

In this paper, we present a method for a quantitative evaluation of the parameters of the deep acceptors (detrapping time  $\tau$ , concentration  $N_T$  and energy level  $E_T - E_V$ ) of a Schottky-type CdTe detector through the measurement of Schottky barrier lowering, on the basis of the conventional model of charge accumulation. Additionally, we modified the charge accumulation model by taking account of the occupation state of the deep acceptor level. The time that the width of a depletion layer becomes smaller than the detector thickness was compared with the time that a photopeak shift occurs in energy spectra.

### 2. Theory

#### 2.1 Extraction of Schottky barrier lowering

The forward current–voltage ( $I_F$ – $V_F$ ) characteristics of Schottky diodes are expressed by the familiar relationship<sup>16)</sup>

$$I_F = I_0 \left[ \exp\left(\frac{qV_F}{kT}\right) - 1 \right] \quad (1)$$

$$\simeq I_0 \exp\left(\frac{qV_F}{kT}\right) \quad \text{for } V_F > 3kT/q, \quad (2)$$

$$I_0 = AA^*T^2 \exp\left(-\frac{q\phi_{b0}}{kT}\right), \quad (3)$$

$$A^* = \frac{4\pi qm^*k^2}{h^3}, \quad (4)$$

where  $I_0$  is the reverse saturation current,  $q$  is the magnitude of electronic charge,  $k$  is the Boltzmann constant,  $T$  is the absolute temperature,  $A$  is the area of contact,  $\phi_{b0}$  is the barrier height at zero bias,  $m^*$  is the hole effective mass ( $0.35m_0$  for p-type CdTe,<sup>17)</sup>  $m_0$  is the free-electron mass),  $A^*$  is the Richardson constant ( $42 \text{ A cm}^{-2} \text{ K}^{-2}$  for p-type CdTe), and  $h$  is Planck's constant. Using eq. (2),  $I_0$  can be determined from the straight line intercept of a semi-log  $I_F$ – $V_F$  plot at  $V_F = 0$ . The value of  $\phi_{b0}$  is obtained from the equation

$$\phi_{b0} = \frac{kT}{q} \ln\left(\frac{AA^*T^2}{I_0}\right). \quad (5)$$

The reverse current of Schottky contact  $I_R$  has a dependence on the reverse bias  $V_R$  due to the effect of Schottky barrier

lowering  $\Delta\phi_b$ .  $I_R$  is expressed as

$$I_R = I_0 \exp\left(\frac{q\Delta\phi_b}{kT}\right). \quad (6)$$

Therefore,  $\Delta\phi_b$  can be obtained using

$$\Delta\phi_b = \frac{kT}{q} \ln\left(\frac{I_R}{I_0}\right). \quad (7)$$

$\Delta\phi_b$  is mainly attributed to the presence of an interfacial layer between metal and semiconductor,<sup>18,19)</sup> and given by

$$\Delta\phi_b \approx \alpha \mathcal{E}_0, \quad (8)$$

where  $\alpha$  is a constant having dimensions of length and  $\mathcal{E}_0$  is the electric-field strength at the Schottky barrier. We can estimate  $\mathcal{E}_0$  by measuring  $\Delta\phi_b$ .

## 2.2 Conventional model of charge accumulation

Many researchers have reported that the behavior of polarization phenomena can be explained by a model which considers the effects of deep acceptors.<sup>9–12)</sup> In general, compound semiconductors contain large numbers of deep acceptors derived from crystal defects. Malm and Martini<sup>9)</sup> consider that the application of a bias voltage to a CdTe detector with a Schottky contact sweeps the holes out of the depletion volume from the deep acceptor level (called detrapping). In this process, deep acceptors negatively ionize with a detrapping time  $\tau$  given by

$$\tau = \frac{1}{N_V \sigma v} \exp\left(\frac{E_T - E_V}{kT}\right), \quad (9)$$

where  $N_V$  is the effective density of states for the valence band,  $\sigma$  is the acceptor capture cross section,  $v$  is the thermal velocity of a hole,  $E_T$  is the energy of the deep acceptor level, and  $E_V$  is the energy of the top of the valence band. When a bias voltage is applied at time  $t = 0$ , the concentration of ionized deep acceptors  $N^-$  will change as

$$N^-(t) = N_T \left[1 - \exp\left(-\frac{t}{\tau}\right)\right], \quad (10)$$

where  $N_T$  is the total concentration of deep acceptors. The accumulation of ionized deep acceptors changes the electric field  $\mathcal{E}$  in the depletion region. We assume that the deep acceptors are uniformly distributed throughout the CdTe bulk. In the conventional model of charge accumulation, whole deep acceptors begin to detrapp with a detrapping time  $\tau$  after applying the bias voltage. When the bias voltage is applied at time  $t = 0$ , the potential  $V$  will be obtained by solving Poisson's equation

$$\frac{d^2}{dx^2} V(x, t) = \frac{qN^-(t)}{\epsilon} = \frac{qN_T}{\epsilon} \left[1 - \exp\left(-\frac{t}{\tau}\right)\right], \quad (11)$$

where  $x$  is the position from the interface of the Schottky contact,  $\epsilon$  is the dielectric constant of the semiconductor ( $\epsilon = 10.2\epsilon_0$  for CdTe,<sup>20)</sup> where  $\epsilon_0$  is the dielectric constant in vacuum). When the bulk is fully depleted, the distribution of the electric field  $\mathcal{E}(x, t)$  in the CdTe bulk is obtained with the boundary conditions of  $V(0) = V_R$  and  $V(D) = 0$ .

$$\mathcal{E}(x, t) = -\frac{dV}{dx}$$

$$= \frac{qN_T}{2\epsilon} \left[1 - \exp\left(-\frac{t}{\tau}\right)\right] (D - 2x) + \frac{V_R}{D}, \quad (12)$$

where  $D$  is the thickness of the CdTe bulk and  $V_R$  is the reverse bias voltage. It should be noted that when the depletion width becomes smaller than the thickness of the bulk,  $D$  in eq. (12) must be replaced by the depletion width  $W$  given by

$$W(t) = \sqrt{\frac{2\epsilon V_R}{qN_T[1 - \exp(-t/\tau)]}} \leq D. \quad (13)$$

The electric-field strength at the Schottky contact  $\mathcal{E}_0$  is given by

$$\begin{aligned} \mathcal{E}_0(t) &= \mathcal{E}_0(t) = \frac{\Delta\phi_b(t)}{\alpha} \\ &= \frac{qN_T}{2\epsilon} \left[1 - \exp\left(-\frac{t}{\tau}\right)\right] D + \frac{V_R}{D}. \end{aligned} \quad (14)$$

At  $t = 0$  and  $\infty$ ,  $\mathcal{E}_0$  becomes

$$\mathcal{E}_0(0) = \frac{\Delta\phi_b(0)}{\alpha} = \frac{V_R}{D}, \quad (15)$$

$$\mathcal{E}_0(\infty) = \frac{\Delta\phi_b(\infty)}{\alpha} = \sqrt{\frac{2qN_TV_R}{\epsilon}}. \quad (16)$$

From eqs. (15) and (16), the constant  $\alpha$  and  $N_T$  are obtained by measuring  $\Delta\phi_b$  at  $t = 0$  and  $\infty$ .  $\tau$  is obtained from the slope of  $\Delta\phi_b(t)$  with time at  $t = 0$ .

$$\tau = \frac{qN_T}{2\epsilon} \alpha D \left( \frac{d\Delta\phi_b(t)}{dt} \bigg|_{t=0} \right)^{-1}. \quad (17)$$

Furthermore, the depth of the deep acceptor level  $E_T - E_V$  can be determined by measuring the temperature dependence of  $\tau$  using eq. (9).

## 3. Experimental Procedure

The CdTe single crystal used in this study was grown by the traveling heater method (THM) of Acrorad Co., Ltd. It has a p-type resistivity of  $\sim 10^9 \Omega \text{ cm}$ .<sup>5)</sup> CdTe wafer was mechanically polished and then chemically etched with Br-methanol solution. A sample with an area of  $8 \times 8 \text{ mm}^2$  and thickness of 0.5 mm was cut from the wafer. The sample was cleaned using trichloroethylene, acetone, and deionized water. To make a stoichiometric CdTe surface, He plasma treatment was carried out by the inductively coupled plasma (ICP) method. Details of the plasma treatment are described in our previous papers.<sup>21,22)</sup> An Al Schottky contact was formed on the CdTe(111)Te face by electron beam evaporation at a pressure of  $\sim 10^{-6} \text{ Pa}$ . The thickness of the Al layer was approximately 50 nm. The substrate temperature during the evaporation was room temperature. On the opposite side, a Pt ohmic contact was formed by electroless plating. Finally, the sample was cut to an area of  $3 \times 3 \text{ mm}^2$  and thickness of 0.5 mm, and mounted on an alumina plate.

The current-voltage characteristics of the fabricated detector were measured in a shield box (Advantest R12706A) using an electrometer (Keithley 6517A) at 0, 10, 20, 30, 40, and 50 °C. The spectrometric performance of the detector was evaluated at 20 °C. The detector was irradiated with a <sup>241</sup>Am  $\gamma$ -ray source from the cathode (Pt contact) side. The measuring system consisted of a shaping

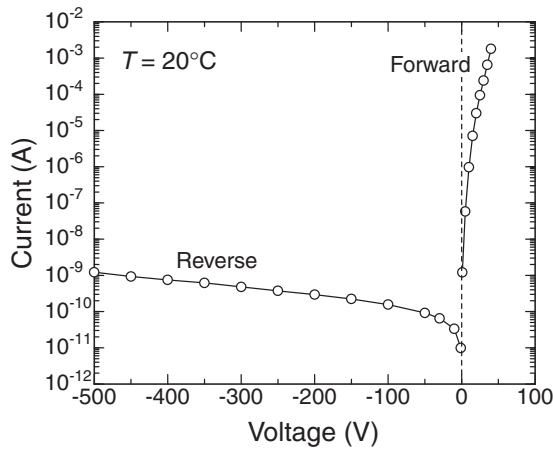


Fig. 1. Current–voltage characteristics of CdTe detector. The measurement was performed at 20°C. The barrier height at zero bias is extracted to be 0.973 eV.

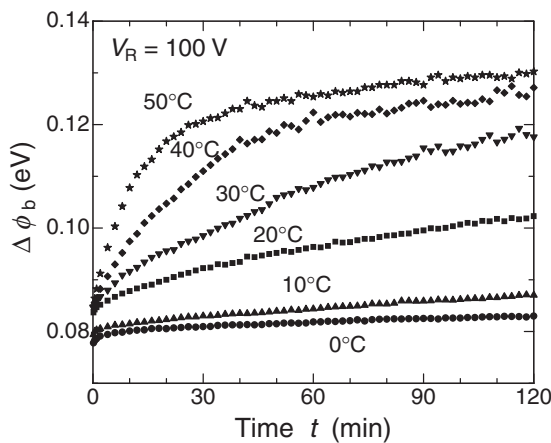


Fig. 2. Time dependence of  $\Delta\phi_b$  measured at each temperature. The applied bias voltage is constant at 100 V.

amplifier (Clear-Pulse 4417), a charge sensitive amplifier (Clear-Pulse 5102), and a pulse height analyzer (Clear-Pulse 1217). The shaping time was set at 1  $\mu$ s. During the measurements, temperature was maintained constant within  $\pm 0.2^\circ\text{C}$  using a thermostatic chamber and a temperature controller.

## 4. Results and Discussion

### 4.1 Electric-field distribution based on conventional model

Figure 1 shows the current–voltage characteristics of the fabricated CdTe detector. It is observed that a rectification property is obtained, which indicates that a Schottky contact is produced between the Al and CdTe surface. From eq. (5), the barrier height at zero bias is found to be 0.973 eV.

Figure 2 shows the time dependence of  $\Delta\phi_b$  measured at each temperature. The applied bias voltage is constant at 100 V. As shown in this figure,  $\Delta\phi_b$  increases with time, and the rate of increase of  $\Delta\phi_b$  becomes higher as the temperature increases. At 40 and 50°C, the value of  $\Delta\phi_b$  gradually saturates within the measuring time. These results seem to obey eqs. (9) and (10).  $\Delta\phi_b$  at  $t = 0$  is estimated to be 0.0850 eV from the data of 50°C, and the constant  $\alpha$  is determined to be 425 nm from eq. (15). The saturation value  $\Delta\phi_b(\infty)$  is determined to be 0.1298 eV from the value after

Table I. Detrapping time  $\tau$  at each temperature for two models.

| Temperature<br>(°C) | Detrapping time $\tau$ (min) |          |
|---------------------|------------------------------|----------|
|                     | Conventional                 | Modified |
| 0                   | 1719                         | 4695     |
| 10                  | 944                          | 2669     |
| 20                  | 260                          | 836      |
| 30                  | 99.4                         | 351      |
| 40                  | 40.9                         | 149      |
| 50                  | 16.2                         | 56.6     |

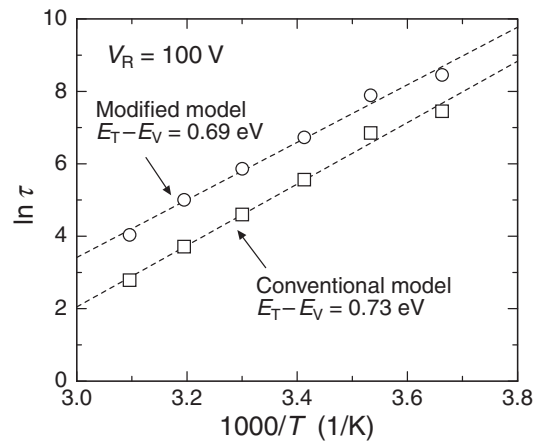


Fig. 3.  $\ln \tau$  vs  $1/T$  plots for two models.

5 h at 50°C. Using eq. (15),  $N_T$  is determined to be  $2.63 \times 10^{11} \text{ cm}^{-3}$ . The values of  $\tau$  are determined using eq. (17) and are listed in Table I. It is found that  $\tau$  is increased as temperature is decreased, in agreement with eq. (9).

The energy level  $E_T - E_V$  can be calculated from the temperature dependence of  $\tau$ . Figure 3 shows  $\ln \tau$  vs  $1/T$  plots, which give a value of 0.73 eV. We consider that the obtained energy level arises from the cadmium vacancies, as reported by Vul *et al.* at  $E_V + 0.77 \text{ eV}$ <sup>23)</sup> (from electro-absorption spectra) and Ayoub *et al.* at  $E_V + 0.6 \text{ eV}$ <sup>24)</sup> (from photoinduced transient current spectroscopy). Ayoub *et al.* also described that this level can be responsible for polarization phenomena.<sup>24)</sup>

Figure 4 shows the time dependence of the distribution of electric field in the bulk of the CdTe detector calculated from eq. (12), using  $V_R = 100 \text{ V}$  and the extracted parameters  $N_T$  and  $\tau$ . The position  $x$  is measured in a direction perpendicular to the Schottky contact. As shown in Fig. 4, the distribution of the electric field is initially constant throughout the CdTe bulk, but begins to incline progressively with time after applying the bias voltage. The electric-field strength at the Schottky contact increases with time, which leads to an increase in Schottky barrier lowering. As a result, leakage current increases with time, and thus the energy resolution of the detector deteriorates. This effect agrees with actual observed spectra. On the other hand, a photopeak shift in the spectra occurs when the depletion width in CdTe bulk becomes less than the thickness of the bulk.<sup>9)</sup> However, the calculated distribution of the electric field indicates that the bulk is never undepleted. This means that the detector shows no photopeak shift, which is inconsistent with the obtained spectra.

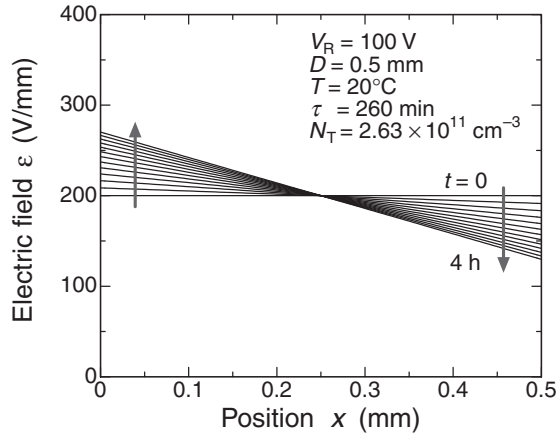


Fig. 4. Time dependence of distribution of electric field in bulk of CdTe detector calculated from eq. (12), using  $V_R = 100$  V,  $N_T = 2.63 \times 10^{11}$   $\text{cm}^{-3}$  and  $\tau = 260$  min.  $x = 0$  and  $0.5$  mm correspond to the anode and cathode of the detector, respectively.

This discrepancy is considered to be due to the inaccurate parameters of the deep acceptors and, therefore, an incorrect charge accumulation model. In the next subsection, we modify the charge accumulation model in consideration of the occupation state of the deep acceptor level.

#### 4.2 Modification of charge accumulation model

In the conventional model, it is assumed that all the deep acceptor levels are empty before applying the bias voltage, and will be occupied by electrons after applying the bias voltage. However, the occupation of the deep levels is actually determined by the Fermi–Dirac function. Assuming the zero-temperature approximation, a deep acceptor level is empty if it lies above the Fermi level  $E_F$  and will be occupied by electrons below  $E_F$  [Fig. 5(a)]. For the case of p-type CdTe, a deep acceptor level can cross the Fermi level

when the bands are bent by the formation of a Schottky contact. The intersection point between a deep acceptor level and the Fermi level is symbolized by  $\lambda$ . If the barrier height at zero bias  $\phi_{b0}$  is larger than a deep acceptor level in the vicinity of the Schottky contact, the deep acceptor level is supposed to be situated below  $E_F$ . From our result,  $\phi_{b0} = 0.973$  eV is larger than a deep acceptor level, implying that the deep levels in the region of  $\lambda$  are already occupied before the application of the bias voltage. If a reverse bias is applied, the Fermi level is split into quasi-Fermi levels for electrons and holes. As shown by Rhoderick and Williams,<sup>18)</sup> it is a very good approximation to assume that the hole quasi-Fermi level  $E_{Fp}$  is horizontal throughout the depletion region and coincides with the Fermi level in the semiconductor, while the electron quasi-Fermi level  $E_{Fe}$  is also horizontal and coincides with the Fermi level in the metal [Fig. 5(b)]. When  $E_{Fp}$  is greater than the deep acceptor levels, they begin to be occupied at a rate with a detrapping time  $\tau$ , and negative space charge is accumulated.

According to this model, the potential  $V$  and the electric field  $\mathcal{E}$  must be obtained by solving Poisson's equations in two regions.

$$\frac{d^2}{dx^2} V(x, t) = \begin{cases} \frac{q}{\epsilon} N_T, & 0 \leq x \leq \lambda \\ \frac{q}{\epsilon} N_T \left[ 1 - \exp\left(-\frac{t}{\tau}\right) \right], & \lambda \leq x \leq D. \end{cases} \quad (18)$$

Using the boundary conditions

$$\begin{aligned} \mathcal{E}(\lambda - 0, t) &= \mathcal{E}(\lambda + 0, t), \\ V(\lambda - 0, t) &= V(\lambda + 0, t), \\ V(0, t) &= V_R, \\ V(D, t) &= 0, \end{aligned} \quad (19)$$

the electric field is given by

$$\mathcal{E}(x, t) = \begin{cases} -\frac{qN_T}{\epsilon}x + \frac{qN_T}{2\epsilon D} \left[ D^2 - (D - \lambda)^2 \exp\left(-\frac{t}{\tau}\right) \right] + \frac{V_R}{D}, & 0 \leq x \leq \lambda \\ -\frac{qN_T}{\epsilon} \left[ 1 - \exp\left(-\frac{t}{\tau}\right) \right] x + \frac{qN_T}{2\epsilon D} \left[ D^2 - (D^2 + \lambda^2) \exp\left(-\frac{t}{\tau}\right) \right] + \frac{V_R}{D}, & \lambda \leq x \leq D. \end{cases} \quad (20)$$

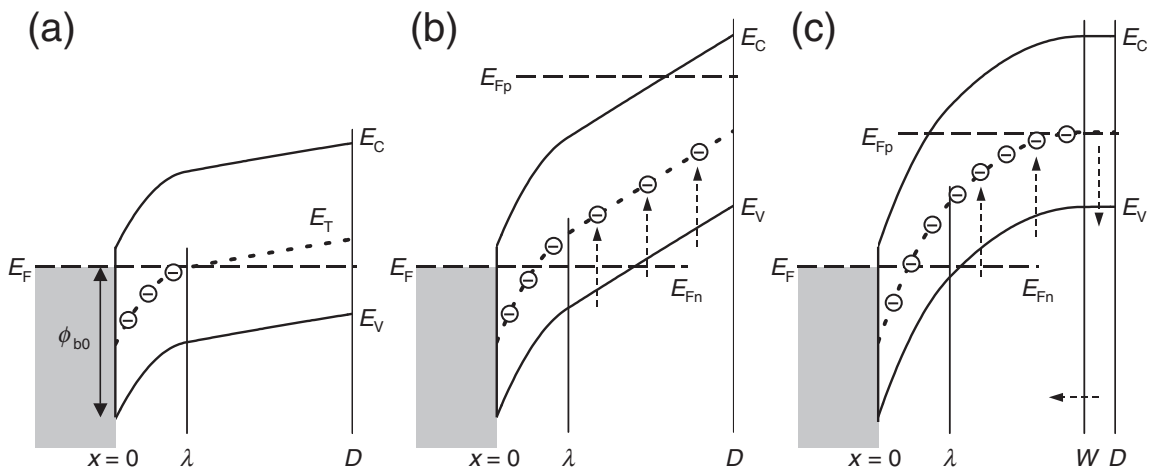
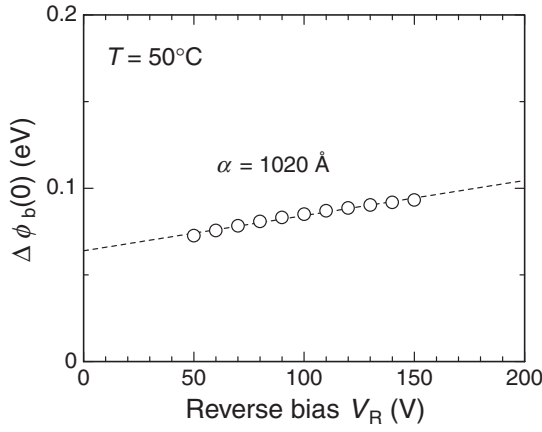


Fig. 5. Energy band diagram of Schottky-type CdTe detector taking account of deep acceptor level. (a) Zero bias, (b) after biasing, and (c) diminishing depletion width after biasing.

Fig. 6.  $\Delta\phi_b(0)$  vs  $V_R$ .

In this case, the electric-field strength at the Schottky contact  $\mathcal{E}_0$  is given by

$$\begin{aligned}\mathcal{E}(0, t) &= \mathcal{E}_0(t) = \frac{\Delta\phi_b(t)}{\alpha} \\ &= \frac{qN_T}{2\epsilon D} \left[ D^2 - (D - \lambda)^2 \exp\left(-\frac{t}{\tau}\right) \right] + \frac{V_R}{D}.\end{aligned}\quad (21)$$

The depletion width is also modified as

$$W(t) = \sqrt{\frac{2\epsilon V_R - qN_T \lambda^2 \exp(-t/\tau)}{qN_T [1 - \exp(-t/\tau)]}} \leq D. \quad (22)$$

At  $t = 0$ ,  $\mathcal{E}_0$  becomes

$$\mathcal{E}_0(0) = \frac{\Delta\phi_b(0)}{\alpha} = \frac{qN_T}{2\epsilon D} [D^2 - (D - \lambda)^2] + \frac{V_R}{D}. \quad (23)$$

Using eq. (23), the constant  $\alpha$  is obtained from the straight-line slope of a  $\Delta\phi_b(0)$ – $V_R$  plot. Figure 6 shows the  $\Delta\phi_b(0)$ – $V_R$  plot, indicating that  $\alpha = 102$  nm. An expression of  $\mathcal{E}_0$  at  $t = \infty$  is the same as eq. (16) and gives  $N_T = 4.56 \times 10^{12} \text{ cm}^{-3}$ .  $\alpha$  is lower by a factor of 4.2, and  $N_T$  is higher by a factor of 17 than those of the conventional model. The straight-line intercept of the  $\Delta\phi_b(0)$ – $V_R$  plot yields  $\lambda$ . From Fig. 6,  $\lambda$  is determined to be 0.087 mm. Equation (17) is modified as

$$\tau = \frac{qN_T}{2\epsilon} \frac{\alpha(D - \lambda)^2}{D} \left( \frac{d\Delta\phi_b(t)}{dt} \bigg|_{t=0} \right)^{-1}. \quad (24)$$

The values of  $\tau$  are determined from eq. (24) and are listed in Table I. The detrapping times in the modified model are 3–4 times longer than those in the conventional model. The  $\ln \tau$  versus  $1/T$  plots in the modified model are shown in Fig. 3, which give  $E_T - E_V = 0.69 \text{ eV}$ .

Figure 7 shows the time dependence of the distribution of electric field calculated from eq. (20), using  $V_R = 100 \text{ V}$  and modified parameters  $\tau$ ,  $N_T$  and  $\lambda$ . As can be seen from the figure, the electric field at  $t = 0$  rises in the region  $0 \leq x \leq \lambda$ , while it is constant in the region  $\lambda \leq x \leq D$ . As time passes, the distribution of the electric field shifts vertically in the region  $0 \leq x \leq \lambda$ , while it inclines in the region  $\lambda \leq x \leq D$ . The increase in electric-field strength at the Schottky contact with time causes a deterioration of energy resolution. On the other hand, the electric field at the cathode decreases with time and reaches zero at  $t = 61 \text{ min}$ , suggesting that the

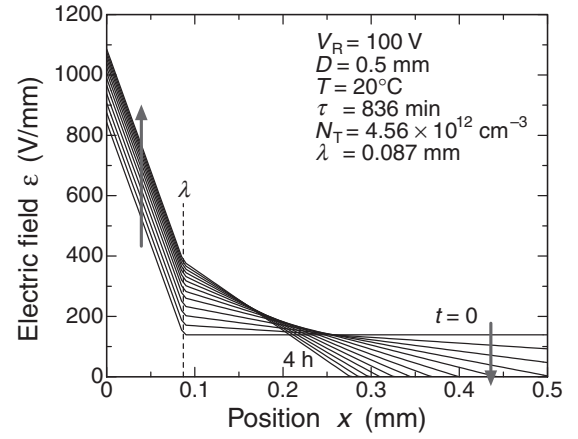


Fig. 7. Time dependence of distribution of electric field in bulk of CdTe detector calculated from eq. (20), using  $V_R = 100 \text{ V}$ ,  $N_T = 4.56 \times 10^{12} \text{ cm}^{-3}$ ,  $\tau = 836 \text{ min}$  and  $\lambda = 0.087 \text{ mm}$ . The positions of  $x = 0$  and  $0.5 \text{ mm}$  correspond to the anode and cathode of the detector, respectively.

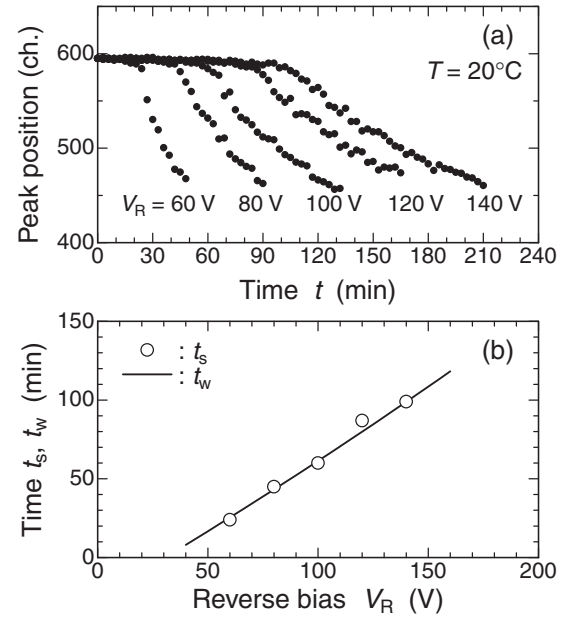


Fig. 8. (a) Time dependence of position of 59.5 keV photopeak observed in radiation measurements at  $20^\circ\text{C}$ . (b) Time that photopeak position begins to shift  $t_s$  and depletion width begins to diminish  $t_w$  against  $V_R$ .

width of the depletion layer begins to diminish. The energy band diagram in which the depletion width becomes less than the thickness of the bulk is illustrated in Fig. 5(c). Holes near the cathode side are trapped after that time and an undepleted region is formed.

It is considered that the shift of photopeak position in radiation measurements is due to the decrease in depletion width. We compared the time  $t_s$  that the photopeak position begins to shift and the time  $t_w$  that the depletion width begins to diminish. Figure 8(a) shows the time dependence of the position of the 59.5 keV photopeak observed in radiation measurements. It was found that the photopeak position begins to shift at a certain time  $t_s$ , and that  $t_s$  becomes longer as the bias voltage increases, as shown in Fig. 8(b). On the other hand,  $t_w$  is calculated from eq. (22) and shown in Fig. 8(b). It is clear that  $t_s$  closely fits the calculated curve. This fact provides evidence for the validity of the modified



model of charge accumulation and the resulting distribution of electric field. Our method enables the simple evaluation of the parameters of deep acceptors as we only have to measure the current–voltage characteristics of a detector.

## 5. Conclusions

We investigated polarization phenomena in a Schottky-type CdTe radiation detector through the measurement of Schottky barrier lowering. On the basis of the conventional model of charge accumulation, the parameters of deep acceptors such as detrapping time  $\tau$ , concentration  $N_T$ , and the depth of the energy level  $E_T - E_V$  were extracted. It is considered that the obtained energy level arises from cadmium vacancies in the crystal. However, the simulated distribution of electric field shows that the CdTe bulk will never be undepleted, which is inconsistent with the observation of photopeak shifts in spectra. On the other hand, the modified model corrects the parameters to  $\tau = 836$  min,  $N_T = 4.56 \times 10^{12} \text{ cm}^{-3}$ , and  $E_T - E_V = 0.69$  eV. In the modified model, the distribution of electric field suggests that the depletion width begins to diminish at a certain time. It was found that the time that the depletion width becomes less than the thickness of the bulk coincides with the time that the photopeak position begins to shift.

Our results show that the modified model of charge accumulation represents an accurate distribution of electric field in CdTe bulk. In addition, our method for the evaluation of deep acceptors is simple as we only have to measure the current–voltage characteristics of a detector. We believe that this method will become useful in the fields of crystal growth and device fabrication not only for CdTe but also other compound semiconductors.

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